Multi-Class Oil Spill Segmentation in SAR Images Using DeepLabV3+ with Progressive Optimization

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Abstract. Oil spills pose a serious threat to the marine ecosystem and require reliable monitoring. Synthetic Aperture Radar (SAR) imagery enables continuous oil spill monitoring under all weather and day-night supervision, but accurate segmentation remains a challenge due to speckle noise, class imbalance, and look-alike phenomena. In this work, a deep learning architecture for oil spill segmentation utilizing Sentinel-1 SAR pictures are gradually refined. To improve contextual feature learning, a DeepLabV3+ model is methodically tuned via progressive loss-function assessment and backbone selection. To specifically handle SAR-specific issues, such as minority-class localization and false alarms from look-alikes.

The suggested system combines an encoder-decoder architecture with an improved Xception backbone and adds a combined Dice + Categorical Cross-Entropy (CCE) loss function. With an Oil Precision of 0.9334, an Oil Recall of 0.9012, an Oil F1-score of 0.9170, an Oil-Spill IoU of 0.8467, and a Mean IoU of 0.8421, the experimental findings show that the improved DeepLabV3+ model outperforms U-Net and previous related studies.

Keywords: Marine Pollution Monitoring, DeepLabV3+, Joint-Loss Functions, Oil Spill Segmentation, Synthetic Aperture Radar (SAR) imagery.

1 Introduction

Over 71% of Earth's surface is covered by water, with oceans covering the majority [1]. Oceans not only provide the habitat for diverse marine biology, but also provide resources for human beings. However, it's constantly threatened by pollution. In the last 50 years alone, 5.86 million tons of oil have escaped from seaborne vessels [2]. Crude oil contains a lot of heavy metals and hazardous chemicals that quickly flow down the food chain and harm the marine environment [3]. A large amount of toxic compounds and heavy metals are present in crude oil, which rapidly enter the marine ecological cycle through the food chain and damage the health of the marine ecosystem once an oil spill occurs [3]. Therefore, it is crucial to identify oil spills in an accurate and timely manner for the purpose of initiating the oil-spill cleanup procedure.

The collection of datasets and simultaneous improvements in computational capacity have fueled the progress of deep learning technologies [4]. For this purpose, satellite imagery can be a valuable tool. Synthetic Aperture Radar (SAR), one of the remote sensing technologies, has proven useful in locating oil leaks regardless of weather conditions or direct sunlight. They are ideal for continual ocean monitoring [5]. The poor back-scatter response in comparison to the surrounding clean sea region causes oil leaks to appear as dark patches in SAR imaging [6]. Crude oils are usually brown to black spots when floating on the seafloor in thick slicks (>3 μm).

Despite the advantages of SAR technology, oil spill detection remains challenging. The dark patches in SAR imagery can also be caused by low-wind areas, biogenic films, or other lookalike phenomena, leading to false detection of oil [7]. In addition to this, SAR images are affected by speckle noise and complex sea surface conditions, further complicating the result [8]. Numerous methods concentrate on either architectural enhancements or loss-function design separately, without methodically optimizing the entire segmentation pipeline under class imbalance and look-alike conditions. Consequently, it is still difficult to segregate oil spills in real-world SAR imaging using many classes.

This paper suggests a progressively tuned deep learning architecture for multi-class oil spill segmentation utilizing Sentinel-SAR images in order to overcome these constraints. To find the best setup, we begin with baseline architectures and then methodically assess segmentation models, encoder backbones, and a joint-loss function approach.

2 Dataset

SAR images from the Sentinel-1 satellites for oil spill segmentation were compiled as a benchmark dataset [9, 10]. Originally, the images were a part of the European Space Agency's Copernicus Open Access Hub, while oil pollution events were verified through records provided by the European Maritime Safety Agency (EMSA) CleanSeaNet service [11]. As Fig. 1 shows, the dataset comprises single polarization channel (VV channel) imagery. All images are resized to a spatial resolution of 1250 × 650 pixels.

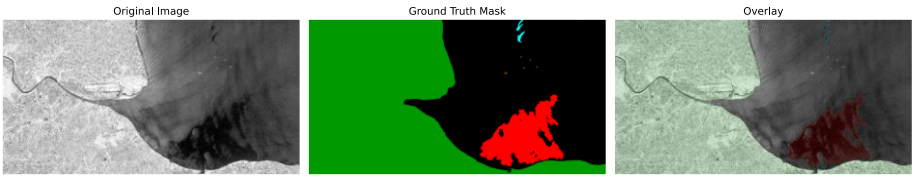

Fig. 1. Image from the dataset with the corresponding mask containing all 5 classes. Oil-Spill: Cyan, Ship: Brown, Look-Alike: Red, Background: Black, Land: Green. Left: Input SAR Image, Middle: Ground Truth Mask, Right: Mask Overlayed on Input Image.

The dataset contains 1112 SAR images, a training set containing 1002 images, and a testing set containing 110. Due to the nature of oil spills, the dataset exhibits class imbalance with the background (sea-surface) class dominating most of the dataset. Oil spills, ships and look-alikes occupy very small regions throughout.

3 Data Preprocessing

Each image was split into overlapping patches of size 512 × 512 with a stride of 256 pixels to quicken training, but at the same time preserving spatial information. For each patch, the image was first normalised. The mask from Fig. 1 (middle) underwent a mapping function where colours were mapped to actual class labels ∈ {0,1,2,3,4}. Multiple patches only contained sea-surface, worsening the class imbalance. From Fig. 2, to fix this, patches containing 1% oil-labelled pixels were retained, and the rest were discarded, the 1% figure was chosen because there were instances of oil spills occupying extremely small regions. To assess the presence of speckle noise in the dataset, Equivalent Number of Looks (ENL) was calculated on 50 × 50 sea-surface patches across randomly sampled instances from the dataset [12]. The values suggested that no smoothing filters were needed as speckle noise levels were acceptable.

$$ENL = \frac{\mu^2}{\sigma^2} \tag{1}$$

Data augmentation techniques, including vertical and horizontal flips, 90-degree rotation, and low-variance Gaussian noise, were introduced to improve model robustness. These augmentations were chosen because they still end up producing realistic SAR images that could occur naturally in the dataset. The patch dataset generated was divided into 80% training 10% validation, using the original training set. The 10% testing set was kept as is.

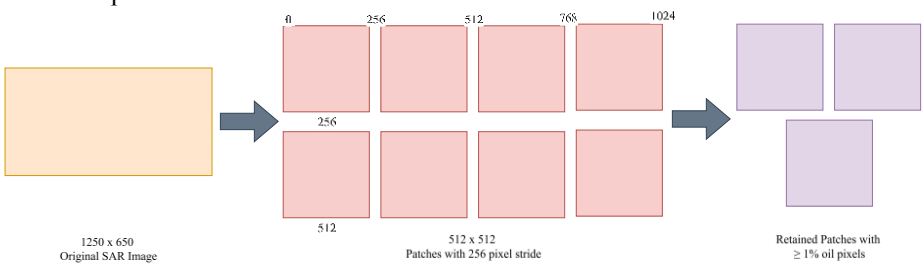

Fig. 2. Patching Strategy Used in Implementation.

4 Architecture Ablation Study

From Fig. 3, the U-Net architecture has an encoder-decoder design with skip connections capturing both high-level and fine-grained spatial features. Encoder creates the latent representation using convolutional and pooling layers, while the decoder

upsamples to help form the predicted mask using the spatial information stored in the bottleneck by transposed convolutions [13].

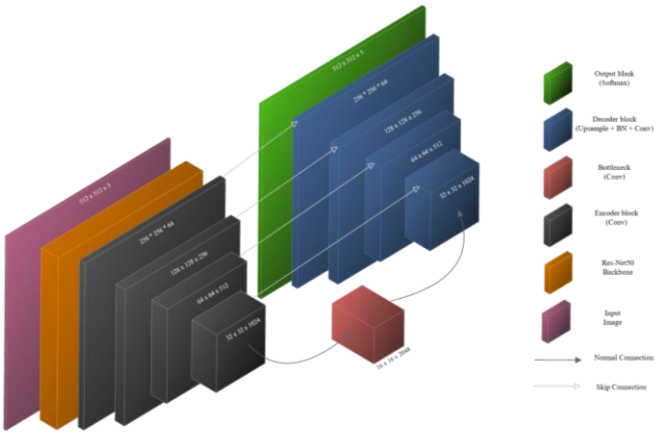

Fig. 3. U-net Architecture Used in Implementation.

As shown in Fig. 4, we used a DeepLabV3+ architecture having an encoder-decoder architecture that has a specialized Atrous Spatial Pyramid Pooling (ASPP) module that processes contextual information at all scales [14]. Parallel atrous convolutions are applied with varying dilation rates by the ASPP module, and global pooling captures features at different scales.

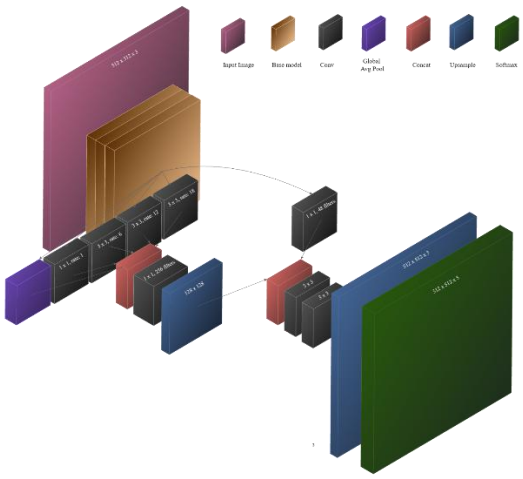

Fig. 4. DeepLabV3+ Architecture Used in Implementation.

From Fig. 5, for choosing the optimal architecture, U-Net and DeepLabV3+ were considered. Both architectures were trained using the same training conditions that included a ResNet-50 backbone with pretrained ImageNet weights as the encoder, a joint loss function combining dice loss and focal loss, a constant learning rate of 0.0001 (1e-4), and Adam optimizer trained over 10 epochs. Class weights used for dice loss throughout this study were unchanged: they were arbitrarily assigned based on the number of pixels as follows: {0.05, 1.0, 3.0, 5.0, 3.0}.

$$L_{Joint} = L_{Dice} + 0.5 \cdot L_{Focal} \tag{2}$$

As Fig. 6 suggests, the U-Net achieved a higher recall and also generalized across classes better with a higher Mean IoU. However, DeepLabV3+ produced more conservative predictions, attaining better precision, F1-score, and IoU on the oil class. False alarms can be caused due to look-alikes in the form of aquatic vegetation, algae, and strong currents, which need to be distinguishable from actual spills. The higher Mean IoU on U-Net was mostly attributed to its better performance on the dominant sea-surface class. Given the study's focus on oil spill detection, DeepLabV3+ made sense to be chosen as the preferred architecture.

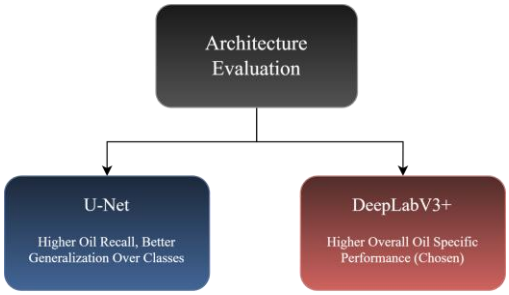

Fig. 5. Choices Tested as Architectures.

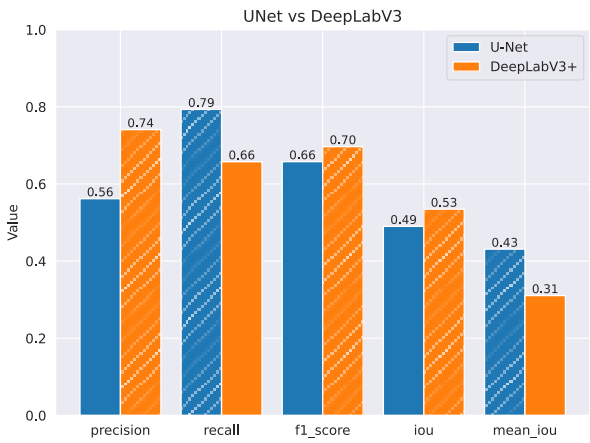

Fig. 6. Metrics calculated using Baseline U-Net and DeepLabV3+ models.

5 Backbone Ablation Study

As shown in Fig. 7, different backbone networks for the encoder part were tested, namely ResNet-50, EfficientNetV2S, and Xception. As Fig. 8 suggests, they were trained under the same training conditions from architecture evaluation, only with a changed number of 30 epochs and the backbone itself. Xception is a variant of the Inception network, which replaces the standard convolutional blocks with depthwise separable convolutions. This decouples spatial and channel-based feature learning, making it efficient [15].

ResNet architecture introduced skip connections that prevented vanishing gradients and incorporated feature reuse [16]. EfficientNetV2S makes use of specialized MBConv and fused-MBConv blocks [17]. It also makes use of compound scaling, enabling faster learning. As Fig. 7 suggests, Xception is the best performing on all fronts, and is therefore selected as the optimal backbone.

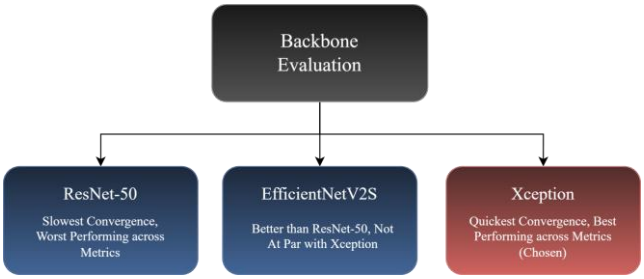

Fig. 7. Choices Tested as Encoder Backbones in DeepLabV3+

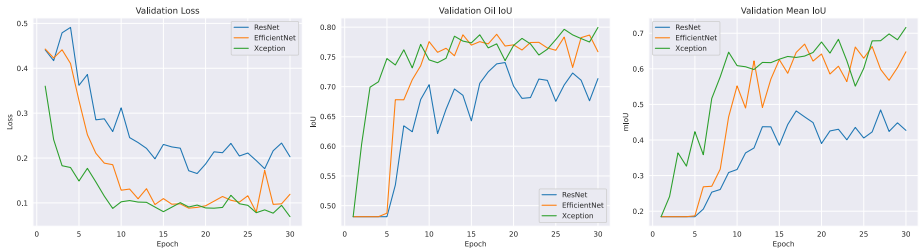

Fig. 8. Loss and Metrics Calculated while Training. Left: Validation Loss, Middle: Validation Oil IoU, Right: Validation Mean IoU.

6 Loss Function Ablation Study

As shown in Fig. 9, Dice, Focal, Categorical Cross-Entropy, and Tversky Loss were first independently trained. All the losses used have been defined as follows:

$$L_{\text{Dice}} = 1 - \frac{2\sum y_i \hat{y}_i + \varepsilon}{\sum y_i + \sum \hat{y}_i + \varepsilon} \tag{3}$$

$$L_{Focal} = -\sum_{c=1}^{C} w_c (1 - \hat{y}_c)^{\gamma} y_c log(\hat{y}_c) \tag{4}$$

$$L_{CCE} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c) \tag{5}$$

$$L_{Tversky} = 1 - \frac{TP + \varepsilon}{TP + \alpha \cdot FN + \beta \cdot FP + \varepsilon} \tag{6}$$

Original Dice + Focal loss was also trained under the same conditions as a baseline comparison. Models were trained up to 30 epochs with early stopping rounds of 10 epochs, and also a callback that lowered the learning rate if the loss was plateauing. As shown in Fig. 10, Dice Loss and Categorical Cross-Entropy were clearly the best performing, they were the only ones able to predict at par with the joint loss. Tversky loss fixated on Oil IoU and wasn't able to generalize well to the other classes. Focal Loss did the exact opposite and was able to generalize well, but didn't do well on Oil IoU.

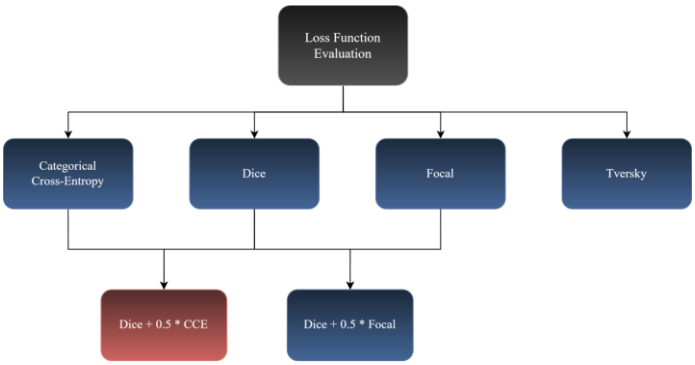

Fig. 9. Choices Tested as Loss Functions.

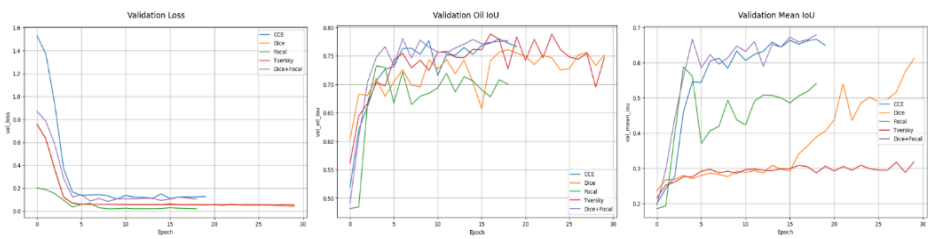

Fig. 10. Loss and Metrics Calculated with Different Losses. Left: Validation Loss, Middle: Validation Oil IoU, Right: Validation Mean IoU.

From Fig. 9, for evaluating model capabilities with joint loss, a new Dice + Categorical Cross-Entropy loss was formulated. As seen in Fig. 11, models with both losses were trained for 200 epochs each. For every chunk of 10 epochs, the best model was saved. The best performing model for Dice + Focal loss was trained for about 190 epochs, and Dice + CCE for 170 epochs. The exact Dice + CCE loss equation used in implementation is as follows:

$$L_{Joint} = L_{Dice} + 0.5 \cdot L_{CCE} \tag{7}$$

Fig. 11. Loss and Metrics Calculated with Dice + Focal and Dice + CCE. Left: Training Loss, Middle: Training Oil IoU, Right: Training Mean IoU.

Apart from the target oil class, there are also other minority classes like Look-alike, Ship, and Land. From Fig. 12, we can observe class-wise IoU for all the classes. Dice + CCE loss outperformed the original loss on all classes except for Look-alike. Oil and Look-alike classes are the most similar; as a result, the IoU of Look-alike is lower for both models compared to the rest of the classes.

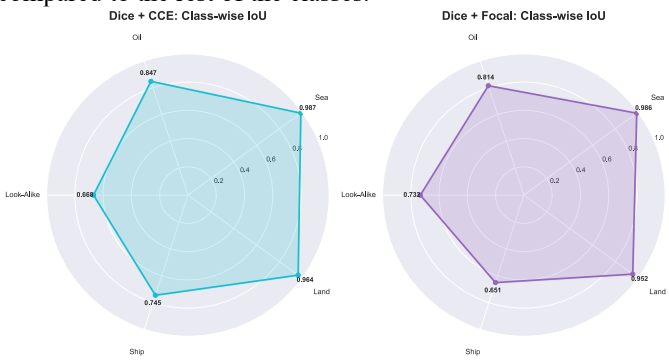

Fig. 12. Class-wise IoU Comparison of Dice + CCE and Dice + Focal Loss Models.

7 Results & Discussion

In this study, we are performing multi-class segmentation, but the primary target is improving performance over the oil-spill class. As a result, all the metrics mentioned from this point forward are oil-spill specific except for Mean Intersection over Union. Here, the metrics are computed using a one-vs-all methodology, where oil-spill class

pixels are considered positives, and the rest of the classes are considered negatives. The metrics used have been defined as follows:

$$Precision = \frac{TP}{TP + FP} \tag{8}$$

$$Recall = \frac{TP}{TP + FN} \tag{9}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{10}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{11}$$

$$mIoU = \frac{1}{C}\sum_{c=1}^{C}\frac{TP_c}{TP_c + FP_c + FN_c} \tag{12}$$

Most pixels belong to the sea surface class, creating the class imbalance. In such cases, a model that simply predicts "sea" for all pixels will achieve high accuracy. This makes accuracy a weak indicator of model performance.

Table 1. Comparison of Our Work with Existing Studies

Ref & Year	Model	Dataset	Metrics - Value
[18] (2025)	DeepLabv3+	Sentinel-1 SAR (Suez Canal)	Oil mIoU: 0.7161, Mean IoU: 0.716
[19] (2025)	U-Net	Krestenitis SAR (Senti-nel-1)	Oil IoU: 0.5806, Mean IoU: 0.6759
[20] (2025)	NOS-Net	Multi-class SAR	Oil IoU: 0.6127, Mean IoU: 0.7029
[21] (2024)	DeepLabv3+ im-proved	Sentinel-1 SAR (Gulf of Mexico)	Mean IoU: 0.8026, F1-score: 0.8866
[22] (2025)	Mask R-CNN	Sentinel-1 SAR	Oil IoU: 0.7480
[23] (2025)	LRA-UNet	Sentinel-1 SAR	Oil mIoU: 0.6736
[24] (2024)	MobileUNet	Sentinel-1 SAR	Oil IoU: 0.7585
Our Model	DeepLabV3+ (Xcep-tion, Dice + CCE Loss)	Sentinel-1 SAR	Oil IoU: 0. 8467, Mean IoU: 0.8421, Oil Precision: 0. 9334, Oil Recall: 0. 9012, Oil F1-score: 0. 9170

Table 1 compares the proposed approach with previously reported SAR-based oil spill detection methods. Similar comparisons are made in Fig. 13 and Fig. 14, where the IoU of existing research is compared with our model's IoU. The DeepLabV3+ model from this work exceeds the existing IoU score, F1-score, and recall; these improvements demonstrate the effectiveness of contextual feature learning and step-wise architecture, backbone, and loss evaluation strategies for SAR oil spill segmentation.

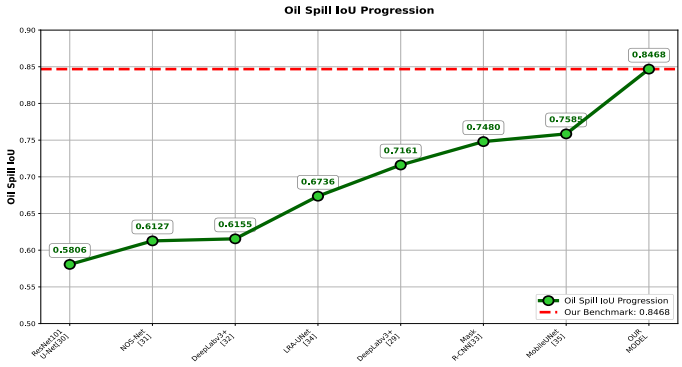

Fig. 13. Visualization of Oil Spill IoU Comparison Between Our Model and Existing Studies.

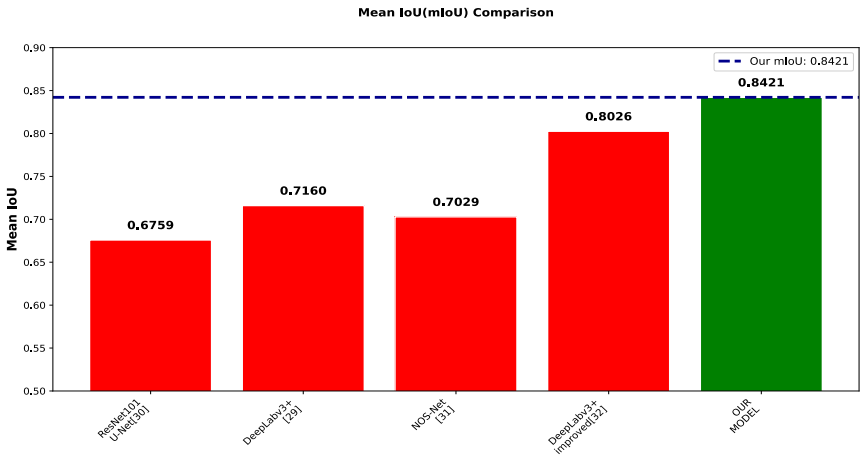

Fig. 14. Visualization of Mean IoU Comparison Between Our Model and Existing Studies.

8 Conclusion

This study presents an optimized deep learning model framework for the automated segmentation of oil spills in oceans using Sentinel-1 SAR imagery. By systematically evaluating the architectural configurations, backbones, and loss functions, it was found

that a DeepLabV3+ model with an Xception backbone and a joint Dice-CCE loss function for evaluation provided the most robust performance.

An integration of a patching strategy was enforced along with the training data formatting, improving the detection accuracy for small-scale spills, making it computationally efficient. Our approach successfully addresses the challenges of SAR imagery analysis, specifically class imbalance and the presence of "look-alikes" (e.g., algae or low-wind zones) by retaining patches of 1% oil-labelled. The model achieved an Oil-Spill IoU of 0.8467 and a Mean IoU of 0.8421, significantly outperforming the previous studies. The model obtained an Oil Precision of 0.9334, an Oil Recall of 0.9012, and an Oil F1-score of 0.9170. An LRA-UNet model was also trained with the same training conditions which didn't perform up to the mark with Mean IoU of 0.4498 and Oil IoU of 0.6944 plateauing in performance after 70 epochs.

Future work will focus on integrating dual-polarization data (VV + VH) and investigating Vision Transformers (ViTs), in order to improve the global contextual understanding of the maritime environment. These developments support more dependable, real-time satellite monitoring, which is crucial for reducing the negative effects of marine oil pollution.

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